Do Now · retrieval

Lesson 084 · A* Search: g, h and f Scores

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is Dijkstra's tentative distance?

Q2

What does f(n) equal in A*?

Q3

What is an admissible heuristic?

Q4

Binary search target12 in sorted [2,4,6,8,10,12,14]. Show midpoint indexes.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — A* orders candidate states using f(n)=g(n)+h(n), where g is known cost from the start and h estimates remaining cost to the goal.

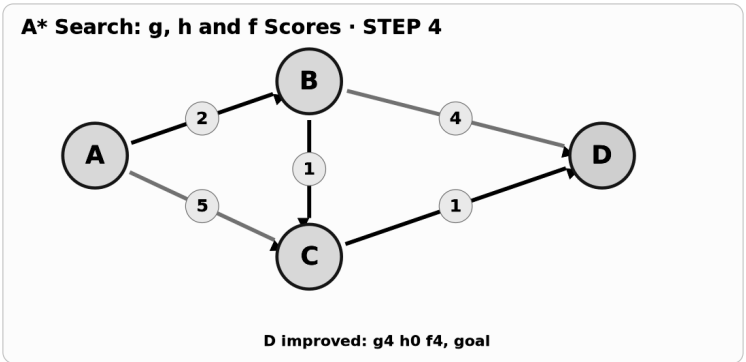
Core idea 2

Apply — A heuristic that never overestimates remaining true cost is admissible; with suitable search implementation/assumptions this supports optimality.

Core idea 3

Evaluate — Track open candidates, their g/h/f and predecessors. An overestimating heuristic or incorrect update can make search miss the shortest path.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

A* — Heuristic-guided shortest-path search.

g(n) — Cost so far from start.

h(n) — Estimated cost to goal.

f(n) — g(n)+h(n) priority.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Find P→R: P→Q = 3, P→R = 8, Q→R = 2. Heuristics: h(P)=5, h(Q)=2, h(R)=0.

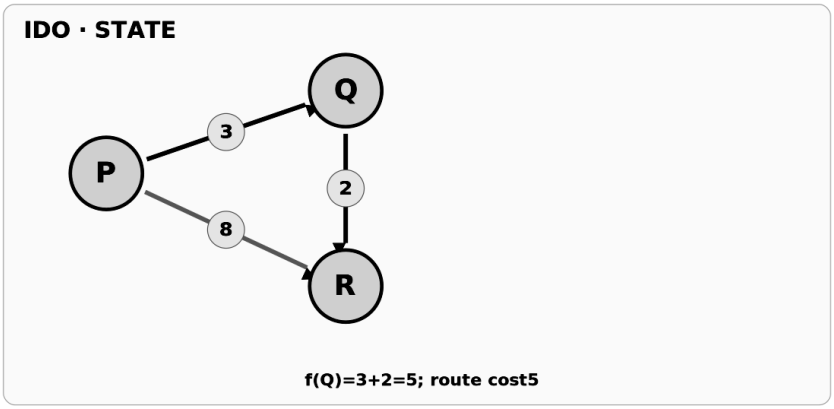
Follow the worked method

1. Calculate all frontier f-scores — From P: Q has g=3,h=2,f=5 ; R has g=8,h=0,f=8 . Put both in OPEN. Choose Q next (smallest f), not simply the nearest-looking node.

2. Improve the path to R — Via Q→R, new g(R)=3+2=5 , which improves the existing 8. Thus f(R)=5+0=5 , and predecessor R becomes Q. OPEN now contains R with the improved values.

3. Confirm the path and heuristic meaning — Select goal R from OPEN, then reconstruct R←Q←P to obtain P→Q→R , cost 5 . The supplied heuristics do not overestimate remaining true cost; the next lesson can explore what happens when this fails.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Compute priorities and choose

Calculate the two g/f pairs from the edges leaving X. State which candidate A* should expand next and why.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

After expanding A, calculate f for B and C using stated graph/heuristics. [6]

Question 2

After expanding B then C, show improvements to g(C) and g(D); state best path. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare A* with Dijkstra when h=0 everywhere, and explain risk of overestimation. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A* h(n) is the exact distance to the goal for every possible heuristic.

Use key vocabulary from memory

1. Explain A* in your own words.

2. Explain g(n) in your own words.

3. Explain h(n) in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.